Probing Attosecond Electron Dynamics using the RABBITT Technique

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For studying ultrafast processes in atoms and molecules, electron dynamics on extremely short timescales must be understood. Attosecond science provides a way to measure these time scales using attosecond pulse trains generated through high harmonic generation [1]. One such technique is the RABBITT (Reconstruction of Attosecond Beating By Interference of Two-photon Transitions), which is widely used to study time delays in the emission of photons. In this method, an electron can reach the same final energy through two different quantum paths involving the absorption and emission of photons. These paths cannot be distinguished, so they interfere with each other and lead to the formation of sidebands in the photoelectron spectrum. The intensity of these sidebands oscillates when the time delay between the extreme ultraviolet (XUV) and infrared (IR) fields is varied. This oscillation provides information on the electron dynamics and time delays in the emission of photons [2]. Such measurements have allowed direct observation of ultrafast electron behavior in atoms, molecules, and solids [3].

In this paper, we present an intuitive overview of the RABBITT technique and highlight its importance in measuring electron dynamics on the attosecond timescale.

I. INTRODUCTION: ATTOSECOND SCIENCE AND HIGH HARMONIC GENERATION

Understanding how electrons move, interact, and transfer energy on extremely short timescales is important for studying many fundamental processes that occur within atoms, molecules, and materials. Without controlling the motion of electrons, modern technology could not exist. The dynamics of such small quantum systems typically occurs on the attosecond time scale ($10^{-18}s$), which is much faster than nuclear motion and most chemical processes. Subpicosecond or femtosecond pulses are used to obtain the resolution of molecular movements. At the attosecond scale, it becomes possible to directly observe and investigate electron motion in atoms, molecules, and solids. Since electrons respond almost instantaneously to external fields, accessing this regime provides valuable insight into fundamental light–matter interactions.

High harmonic generation (HHG), produces the spectrum of a 200 attosecond(as) unipolar pulse that covers the optical spectrum from the near infrared (IR) to the extreme ultraviolet (XUV) regime. The process includes focusing an intense laser pulse on an atomic gas jet, generating higher, equally spaced harmonics of the driving laser frequency, which in the time domain corresponds to a train of attosecond pulses due to the non-linear response of the medium. The strong electric field of the laser distorts the Coulomb potential of the atom, allowing an electron to tunnel ionize. Once ionized, the electron is accelerated by the oscillating laser field and can later recombine with the parent ion. During recombination, the excess kinetic energy of the electron is released in the form of high-energy photons, producing harmonics of the driving laser frequency [1]. This can be understood in three steps: ionization, acceleration, and recombination. Due to the periodic driving laser field, this sequence repeats every half cycle of the laser field, resulting in the emission of a series of odd harmonics. In the frequency domain, this appears as a comb-like spectrum of discrete harmonics, while in the time domain it corresponds to a train of attosecond pulses.

Among the various interferometric techniques developed in attosecond science, the RABBITT (Reconstruction of Attosecond Beating By Interference of Two-photon Transitions) method is widely used to study time delays in photoemission. This technique is based on cross-correlation measurements between the attosecond XUV light field and a phase-locked infrared (IR) laser field. In this technique, an atom is ionized by an XUV pulse in the presence of a weak infrared (IR) field. As a result, additional peaks, known as sidebands, appear in the photoelectron spectrum. These sidebands arise because an electron can reach the same final energy through two different processes involving the absorption and emission of photons.

Since these two processes are indistinguishable, they interfere with each other. This interference leads to an oscillation in the sideband intensity as the time delay between the XUV and IR fields is varied. By analyzing this oscillatory behavior, it is possible to extract phase information and determine time delays associated with the photoemission process [2]. This provides a direct way to study electron dynamics on attosecond timescales.

In this paper, we aim to present a clear and intuitive understanding of the RABBITT technique. The paper is organized as follows: first we discuss attosecond science and HHG, then the RABBITT technique and its calculation, followed by time-delay extraction, applications, and finally experimental considerations and conclusions.

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II. PRINCIPLE OF THE RABBIT TECHNIQUE

The RABBITT method is an interferometric technique that measures time delays in photoemissions to study ultrafast electron dynamics. When an atom is exposed to the XUV attosecond pulse train (ATP), photoelectrons are emitted with discrete energies corresponding to the harmonic frequencies of the driving laser, which are typically odd multiples of the fundamental frequency, leading to a series of peaks in the photoelectron spectrum. When an IR pulse, which is a weak replica of the field used to generate the APT, is introduced with its phase locked to the XUV field, additional peaks, known as sidebands, appear between the main harmonic peaks. These sidebands arise from two-photon transitions (XUV±IR), i.e.,two different quantum pathways through which an electron can reach the same final energy. In the first pathway,the electron absorbs a photon from a lower-order harmonic and simultaneously absorbs an IR photon (XUV+IR). In the second pathway, the electron absorbs a photon from a higher-order harmonic and emits an IR photon (XUV-IR), (FIG. 1). The interference between these two quantum pathways causes the sideband signals to oscillate as the delay between the IR and XUV pulses is varied. These oscillations are the key features of the RABBITT method as they arise due to the phase difference between the two pathways. By measuring the sideband intensity as a function of this delay, one obtains an oscillatory signal whose phase contains information about the photoemission process.Although these processes involve different sequences of photon interactions, they result in the same final energy of the emitted electron.This interference is the central idea behind the RABBITT technique.The relative phase between the two pathways determines whether the interference is constructive or destructive, and this is reflected in the intensity of the sidebands. From the analysis of these oscillations,time delays associated with the electron emissions are extracted.These delays include contributions from the interaction of the electron with the atomic potential as well as the influence of the probing IR field. In this way, the RABBITT technique provides a powerful tool for accessing ultrafast electron dynamics with attosecond resolution [2].

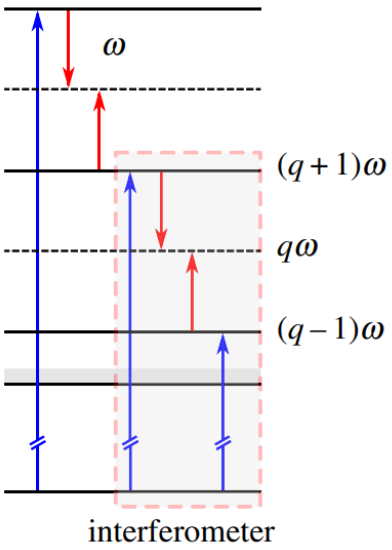


FIG. 1. Each side band q makes up a quantum interferometer, one arm being the absorption of harmonic $q-1$ followed by absorption of an IR photon, and the other the absorption of harmonic $q+1$ followed by emission of an IR photon, leading to thesame final energy.

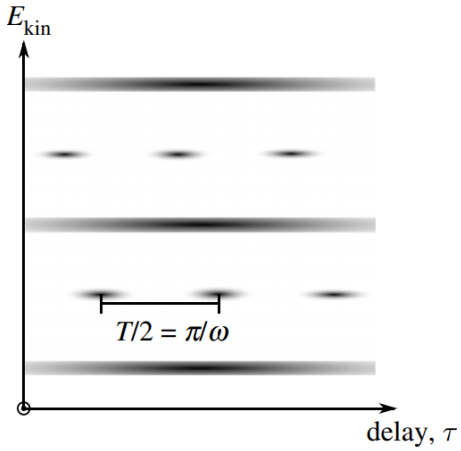


FIG. 2. The intensity of thesideband oscillates as a function of the delay between the pulses, with an oscillation period ofT/2, whereT is the period of the fundamental, and the phase of this oscillation reveals information about the target system.

III. CALCULATION OF SIDEBAND OSCILLATIONS

In the RABBITT technique, the oscillatory behavior of the sideband intensity arises from the interference between two quantum pathways, refer ti figure 3.In this section, we derive this result starting from the electric field and show how the phase of the oscillation can be expressed in terms of a time delay.

The extreme ultraviolet (XUV) field can be written as a superposition of odd harmonics (so here q is an integer, $q = 0, 1, 2...$ of the fundamental frequency ω:

$$E_{\mathrm{XUV}}(t) = \sum_q E_q e^{-i(2q+1)\omega t + i\phi_{2q+1}}, \tag{1}$$

where ϕ_{2q+1} represents the phase of each harmonic.

The infrared (IR) probe field, delayed by a time τ, is given by

$$E_{\mathrm{IR}}(t) = E_0 e^{-i\omega(t-\tau)} + \text{c.c.} \tag{2}$$

We omit the complex conjugate (corresponds to the emission process) for the XUV field because only the ab-

sorption of XUV photons contributes significantly to the ionization process,while for the IR field both absorption and emission processes are essential for producing interference.A sideband at energy $2q\omega$ can be formed through two different pathways.In the first pathway, the electron absorbs a photon from the $(2q-1)$th harmonic and absorbs one IR photon. In the second pathway,the electron absorbs a photon from the $(2q+1)$th harmonic and emits one IR photon.

The corresponding amplitudes can be written as

$$A_{\rm abs} \propto E_{2q-1}\, E_{\rm IR}\, M_{\rm abs}, \quad A_{\rm emi} \propto E_{2q+1}\, E_{\rm IR}^{*}\, M_{\rm emi}. \tag{3}$$

Where Mabs (Memi) are two-photon ionization matrix elements involving absorption (emission) of the fundamental field in the continuum (atomic contributions). Including both field and atomic contributions, the amplitudes can be expressed as

$$A_{\rm abs} = |A_{\rm abs}|e^{i\phi_{\rm abs}}, \qquad A_{\rm emi} = |A_{\rm emi}|e^{i\phi_{\rm emi}}. \tag{4}$$

Substituting the field expressions, the phase factors become

$$A_{\rm abs} \sim e^{-i(2q-1)\omega t + i\phi_{2q-1}}\, e^{-i\omega(t-\tau)}\, e^{i\phi_{\rm atomic}^{(2q-1)}}, \tag{5}$$

$$A_{\rm emi} \sim e^{-i(2q+1)\omega t + i\phi_{2q+1}}\, e^{+i\omega(t-\tau)}\, e^{i\phi_{\rm atomic}^{(2q+1)}}. \tag{6}$$

Simplifying, we obtain

$$A_{\rm abs} \sim e^{-i(2q\omega t - \omega\tau)}\, e^{i\left(\phi_{2q-1}+\phi_{\rm atomic}^{(2q-1)}\right)}, \tag{7}$$

$$A_{\rm emi} \sim e^{-i(2q\omega t + \omega\tau)}\, e^{i\left(\phi_{2q+1}+\phi_{\rm atomic}^{(2q+1)}\right)}. \tag{8}$$

The phase difference between the two pathways is therefore

$$\Delta\phi = \arg(A_{\rm abs}) - \arg(A_{\rm emi}) = 2\omega\tau + (\phi_{2q-1} - \phi_{2q+1}) + \phi_{\rm atomic}, \tag{9}$$

where $\phi_{\rm atomic} = \phi_{\rm atomic}^{(2q+1)} - \phi_{\rm atomic}^{(2q-1)}$ represents the additional phase arising from the photoemission process.

The harmonic phase difference can be interpreted in terms of the group delay of the attosecond pulse train, and the atomic phase corresponds to the intrinsic photoemission delay.These phase contributions can be expressed in terms of time delays by dividing by 2ω:

$$\tau_{\rm XUV} = \frac{\phi_{2q-1} - \phi_{2q+1}}{2\omega}, \qquad \tau_{\rm atomic} = \frac{\phi_{\rm atomic}}{2\omega}. \tag{10}$$

Defining the total delay as

$$\tau_R = \tau_{\rm XUV} + \tau_{\rm atomic}, \tag{11}$$

the phase difference becomes

$$\Delta\phi = 2\omega(\tau - \tau_R). \tag{12}$$

The total amplitude for the sideband is given by

$$A = A_{\rm abs} + A_{\rm emi}, \tag{13}$$

and the corresponding intensity is

$$S_q = |A|^2. \tag{14}$$

Expanding this expression, we obtain

$$S_q = |A_{\rm abs}|^2 + |A_{\rm emi}|^2 + 2|A_{\rm abs}||A_{\rm emi}|\cos\left[2\omega(\tau - \tau_R)\right]. \tag{15}$$

In an actual experiment, the measured signal is obtained by integrating over the spatial interaction region between the XUV and IR fields. The sideband signal can therefore be written as

$$S_q = \int \left(|A_{\rm abs}|^2 + |A_{\rm emi}|^2 + 2|A_{\rm abs}||A_{\rm emi}|\cos\left[2\omega(\tau - \tau_R)\right]\right) d^3r. \tag{16}$$

While spatial variations in the amplitudes do not affect the phase of the oscillation, variations in the delay τ_R across the interaction region can reduce the contrast of the oscillations and affect the accuracy of the extracted time delay [2].

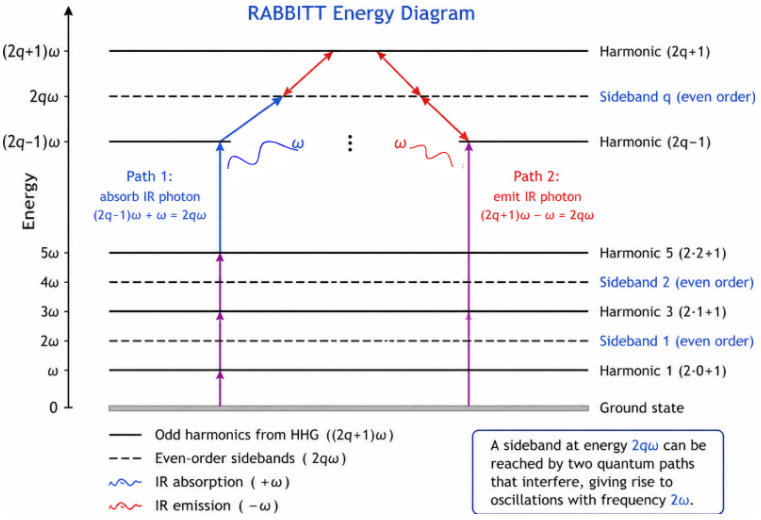


FIG. 3. Energy level diagram of the RABBITT technique

IV. EXTRACTION OF TIME DELAY

The RABBITT technique allows us to measure time delays in photoemission by looking at how the sideband intensity changes with the delay between the XUV and IR pulses. From the previous section, the sideband signal is given by

$$S_q(\tau) = A + B\cos\left[2\omega(\tau - \tau_R)\right], \tag{17}$$

where τ_R is the total delay containing contributions from both the XUV pulse train and the atomic response.

In an experiment, the delay τ is varied in a controlled way using an interferometric setup. For each value of τ, the sideband intensity is measured, giving an oscillating signal as a function of delay. Ideally,this signal would be a perfect cosine curve. However, in real experiments, the data contains noise and small fluctuations.

To extract the time delay, the measured data is fitted to a cosine function of the same form.The phase shift of this fitted curve gives the delay through

$$\tau_R = \frac{\phi}{2\omega}. \tag{18}$$

This procedure is illustrated in Fig. 4(a), where a simulated RABBITT signal is shown. The points represent noisy data, while the smooth curve represents the fitted cosine function. The noise in the data and fluctuations in the delay (often called delay jitter) make the extraction of the phase less straightforward.

In Fig. 4(b), the phase difference between the original (ideal) signal and the fitted curve is shown. This difference corresponds to an error in the extracted delay, denoted by $\Delta\tau$. This means that even though we use a fitting procedure, the extracted delay is not exactly the true value.

To understand how reliable this extraction is, the procedure is repeated many times using simulations, as discussed in [2]. In each simulation, noise and fluctuations are added to the signal, the data is fitted again, and a value of the delay is obtained. This is known as a Monte Carlo approach.

The result of repeating this process many times is shown in Fig. 4(c), where a distribution of extracted delays is obtained. Instead of a single value, we now have a spread of values. The width of this distribution gives the uncertainty in the measurement.

Finally, Fig. 4(d) shows how the precision of the measurement depends on experimental parameters such as the number of data points, the scan length, and the step size. It is observed that using more data points and smaller step sizes improves the accuracy of the extracted delay.

From this analysis, we see that while the time delay can be obtained analytically from the phase of the oscillation, its accurate measurement in experiments depends strongly on noise, delay control, and data sampling. This highlights the importance of careful experimental design in RABBITT measurements.

V. EXPERIMENTAL CONSIDERATIONS AND LIMITATIONS

In real RABBITT experiments, the extraction of time delay is not as ideal as in theory.Several practical factors affect the accuracy of the measurement. As discussed in [2], the spatial and temporal properties of the XUV and IR fields play an important role.

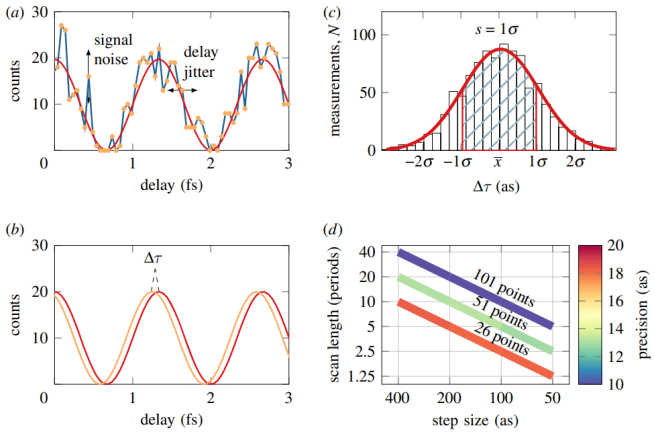


FIG. 4. Illustration of the model used to simulate RABBIT traces

One important factor is the spatial variation of the fields in the interaction region. Since the XUV and IR beams are not perfectly uniform, the delay can vary slightly at different points in space. When the signal is measured, it is effectively averaged over the whole interaction region. This averaging can reduce the contrast of the oscillations and make the phase extraction less accurate.

Another important factor is related to the temporal properties of the pulses. Things like pulse duration, phase stability, and delay jitter can affect the measurement. In particular, small fluctuations in the delay between the XUV and IR pulses can introduce errors in the extracted phase, as also seen in the simulated results.

To improve the precision of the measurement, it is necessary to optimize the experimental parameters. For example, using more data points, choosing a smaller step size while scanning the delay, and ensuring better stability of the laser system can all help in getting a more accurate value of the time delay.

Overall, this shows that even though the theory is quite straightforward, the actual experiment requires careful control of different factors to obtain reliable results.

VI. APPLICATIONS OF THE RABBITT TECHNIQUE

The RABBITT technique is widely used in attosecond science to study ultrafast electron dynamics.One of its main applications is in measuring time delays in photoemission processes. These delays are extremely small (on the order of attoseconds), but they provide important information about how electrons respond to light in atoms and molecules [2].

RABBITT has been used to study time delays in atomic systems, where it helps in understanding how electrons are emitted after absorbing high-energy photons. Early experimental work on attosecond pulse trains

generated through high harmonic generation [1] made such measurements possible. By comparing delays for different electronic states, one can gain insight into the internal structure and dynamics of atoms.

In molecular systems, electron dynamics are more complex due to multiple atoms and bonding effects.RABBITT measurements can be used to study how these factors influence electron emission and provide information about molecular dynamics. Theoretical approaches, such as time-dependent density functional theory, have been used to model such processes and compare with experimental results [3].

The technique is also applied to solid-state systems, where it helps in studying electron emission and transport from surfaces. These studies are important for understanding material properties and ultrafast processes in condensed matter systems [3].

In addition to studying physical systems, RABBITT is also useful for testing theoretical models.The experimentally measured time delays can be compared with theoretical predictions, helping to improve our understanding of light–matter interaction at very short timescales [2].

Overall,the RABBITT technique provides a powerful way to probe electron dynamics in atoms,molecules, and solids, making it an important tool in modern ultrafast physics.

VII. CONCLUSION

In this work, we studied the RABBITT technique as a method for probing ultrafast electron dynamics on the attosecond timescale.Starting from the basic idea of interference between two quantum pathways, we derived the expression for the sideband oscillations and showed how the phase of these oscillations is directly related to the time delay in photoemission.

One important point that becomes clear is that the measurement of time delay is not a direct observation, but rather an indirect one. The delay is obtained from the phase of an oscillating signal, which itself arises due to interference between two indistinguishable processes. This makes the technique conceptually similar to interference experiments in optics, but applied at the level of electron dynamics.

Another key aspect is the role of the atomic phase. From the calculation, it is seen that the total delay contains contributions from both the XUV field and the atomic response. This shows that the measured delay is not purely a property of the atom, but also depends on the characteristics of the light field used in the experiment.

From the experimental side, it is clear that extracting the time delay accurately is not striaghtforward. Factors such as noise, delay jitter, and spatial averaging affect the measured signal. The use of fitting procedures and Monte Carlo simulations, as discussed in [2], helps in understanding the uncertainty and reliability of the extracted delay.

The applications of the RABBITT technique further highlight its importance. It has been used to study electron dynamics in atoms, molecules, and solids, showing that it is a versatile tool in ultrfast physics. At the same time, it also provides a way to test theoretical models by comparing experimental results with simulations.

Overall, this study shows that th RABBITT technique combines concepts from quantum mechanics, optics, and experimental physics. While the basic idea is simple, the actual implementation and interpretation require careful consideration of both theoretical and experimental factors. This makes it an interesting and powerful method for studying processes that occur on extremely short timescales.

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